Comparative Analysis of Data Sampling Techniques for Legal Text Classification in Real-World Scenarios

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Abstract. This article investigates the use of machine learning for classifying legal texts, focusing on the challenges posed by imbalanced class distributions in datasets. In Brazil, where legal processes are numerous and complex, machine learning could improve judicial efficiency and decision-making speed. However, skewed data distributions make it difficult for standard algorithms to perform well across all classes, often neglecting minority categories. To address this, imbalance learning strategies such as oversampling, undersampling, and hybrid methods are applied to balance data sets and improve classification accuracy. Using real legal data from São Paulo, this study evaluates these techniques' effectiveness in binary classification, providing valuable insights into their applicability in legal settings.

Keywords: Machine learning · Text classification · Data sampling · Imbalanced data · Natural language processing.

1 Introduction

Machine learning (ML) techniques have become essential in improving the efficiency of justice systems, especially in Brazil, where a large number of active legal cases require effective management. According to the National Council of Justice's "Justice in Numbers" report for 2022, there are more than 77.23 million active cases [13]. Integrating ML tools for analyzing, classifying, and retrieving legal documents can greatly increase the speed and accuracy of court processes. Text classification, a primary task in Natural Language Processing (NLP), plays a critical role in categorizing legal documents, addressing the high demand and limited resources in the public sector [3].

Despite these benefits, ML algorithms often struggle with imbalanced class distributions in datasets, which can reduce prediction accuracy and lead to biases toward majority classes [11]. This imbalance issue is particularly challenging in legal datasets, where minority class instances are often under-represented but essential for robust decision-making.

To mitigate this issue, researchers have developed imbalance learning techniques aimed at improving the accuracy of the model by modifying prediction algorithms or adjusting datasets [5]. Data sampling strategies, such as oversampling and undersampling, help balance class distributions by either replicating or generating synthetic samples of minority classes or by reducing instances of majority classes.

This article reviews various techniques for managing class imbalance, including the Synthetic Minority Oversampling Technique (SMOTE) and its variations, such as SMOTE-IPF and DSMOTE, which have been shown to improve classification performance metrics such as accuracy, precision, recall, and F-score [4, 14]. Recent advances, such as the OBGAN technique, leverage generative aversarial networks to improve the precision of the minority class by creating synthetic samples near the class boundaries [9].

Using a dataset of lawsuits from the São Paulo Court of Justice, this research evaluates the effectiveness of oversampling, undersampling, and combined methods for binary classification tasks. This study contributes a reproducible assessment of resampling techniques specifically for legal text data, offering recommendations for managing imbalanced datasets in complex domains.

2 Oversampling and Undersampling Methods

This section discusses widely implemented oversampling and undersampling techniques to address class imbalances in datasets. Key methods include Random Oversampling (RO) [1], Synthetic Minority Oversampling Technique (SMOTE) [2], Borderline-SMOTE [6], and Adaptive Synthetic Sampling (ADASYN) [7] for oversampling. For undersampling, the methods discussed are random undersampling (RU), near miss undersampling (NMU) [16], Tomek Links Undersampling (TLU) [8], and the Edited Nearest Neighbours Rule (ENN).

2.1 Oversampling Techniques

Random Oversampling (RO) involves duplicating instances from the minority class to balance the class distribution [1]. Although RO is straightforward, it can lead to overfitting since it merely replicates existing samples without introducing variation. In contrast, **SMOTE** creates synthetic samples by interpolating between instances of the minority class, which effectively enriches the dataset with plausible new data points rather than exact duplicates [2].

Borderline-SMOTE [6] is a targeted variant of SMOTE that focuses on minority class instances near the decision boundary between classes. By generating synthetic samples close to these critical instances, Borderline-SMOTE aims to enhance model performance in challenging regions where the risk of misclassification is higher. **Adaptive Synthetic Sampling (ADASYN)** [7] further refines this process by adaptively generating synthetic samples based on the density of minority class instances in different regions of the feature space.

2.2 Undersampling Techniques

Random Undersampling (RU) reduces the dataset size by randomly removing samples from the majority class, which helps balance the class distribution. However, RU can result in the loss of valuable information, as it discards data points without considering their potential contribution to the model's learning. Alternatively, **Near Miss Undersampling (NMU)** [16] retains majority class samples that are closest to minority class instances based on the k-nearest neighbor algorithm.

Tomek Links Undersampling (TLU) [8] identifies and removes majority class samples that form Tomek links with minority class samples, effectively cleaning up the decision boundary. By removing these specific samples, TLU refines the separation between classes, making classification boundaries more distinct. **Edited Nearest Neighbors (ENN)**, proposed by Dennis Wilson in 1972, further enhances this effect by removing majority class samples that are misclassified by their nearest neighbors.

2.3 Text Vectorization Techniques

Transformation of textual information into numerical vectors is crucial for machine learning algorithms to process text data effectively. The Term Frequency-Inverse Document Frequency (TF-IDF) technique is widely used for this purpose [15].

Implementation of TF-IDF The TF-IDF score is calculated using the term's frequency and the inverse document frequency, which helps to diminish the weight of terms that occur very frequently across the corpus and are, hence, less informative.

$$IDF(t) = \log\left(\frac{1+N}{1+DF(t)}\right) + 1 \tag{1}$$

$$TF - IDF(t,d) = TF(t,d) \times IDF(t) \tag{2}$$

Where N is the total number of documents in the corpus, $DF(t)$ is the number of documents containing the term t, and $TF(t,d)$ is the frequency of term t in document d.

This vectorization process enables the application of machine learning algorithms to perform tasks such as classification, clustering, and information retrieval in NLP applications.

3 Experimental Study

This section delves into a comparative study using a real dataset of lawsuit decisions from the Court of São Paulo. The dataset is bifurcated into two classes: class_1, representing a specific legal theme, and class_0, representing all other cases. We investigate the effectiveness of various data sampling techniques in optimizing the error rate for classifying lawsuit decisions related to the targeted theme.

3.1 Classification Approach

We employ the Support Vector Machines (SVMs) with Stochastic Gradient Descent (SGD) as our classification strategy. SGD offers an efficient and effective means of fitting linear classifiers under convex loss functions, which is particularly advantageous for large-scale and sparse machine learning problems commonly found in text classification and natural language processing [12]. The specific implementation details of the SVM classifier used in this study are summarised in Table 1.

3.2 Procedure

The classification accuracy of the lawsuit decisions is evaluated using eight different data sampling methods. Each method aims to address the imbalance in the training data, either by adjusting the class proportions or by enhancing the representativeness of the minority class.

- Method 1: Unbalanced - uses the training set without any modifications.
- Method 2: Random Undersampling - balances the classes by randomly reducing the majority class.
- Method 3: Near Miss Undersampling - selects samples from the majority class that are closest to the minority class samples.
- Method 4: SMOTE - generates synthetic samples by interpolating between existing minority class samples.

Table 1: SGD Classifier tuned parameters

Parameter	**Value**	**Parameter**	**Value**
alpha	0.0001	n_iter_no_change	5
average	False	n_jobs	-1
class_weight	None	penalty	‘l2‘
early_stopping	False	power_t	0.5
epsilon	0.1	random_state	123
eta0	0.001	shuffle	True
fit_intercept	True	tol	0.001
l1_ratio	0.15	validation_fraction	0.1
learning_rate	‘optimal‘	verbose	0
loss	‘hinge‘	warm_start	False
max_iter	1000		

- Method 5: ADASYN - adapts the number of synthetic samples according to the density of the minority class.
- Method 6: Random Oversampling - Increases the number of minority class samples by duplicating existing samples.
- Method 7: Borderline-SMOTE - focuses on minority samples near the decision boundary.
- Method 8: Edited Nearest Neighbours - removes majority class samples that are misclassified by their nearest neighbors.
- Method 9: Tomek Links Undersampling - removes samples that form Tomek links between classes, thus clarifying the decision boundary.

3.3 Dataset Distribution

The dataset was divided into training, validation, and production sets, with a detailed breakdown of class distribution across these sets to ensure a comprehensive evaluation of the sampling methods employed. It was divided into these phases to ensure model robustness and accuracy. The training set helps the model learn patterns, the validation set optimizes performance by tuning parameters and preventing overfitting, and the production set tests the model's real-world effectiveness on new data. This approach ensures balanced, reliable model performance across all stages. The training set included 13,573 texts, the validation set consisted of 2,560 texts, and the production set contained 350 texts. Each textual entry in the dataset is associated with a specific class label, indicating the nature of the content. The objective of the trained model is to learn linguistic and semantic patterns that distinguish each class, enabling it to correctly classify previously unseen texts. Thus, when presented with a new textual entry outside the training dataset, the model can infer the corresponding class label based on the features learned during training.

3.4 Evaluation Metrics

We assess the performance using metrics such as balance accuracy, precision, recall, F1-score, geometric mean (g-mean), and kappa. The geometric mean combined "sensitivity" and "specificity" metrics into a single score that balances both concerns. Sensitivity refers to the true positive rate and summarizes how well the positive class was predicted. Specificity complements sensitivity, or the true negative rate, and summarises how well the negative class was predicted. Cohen's kappa is calculated based on the confusion matrix. However, in contrast to calculating overall accuracy, Cohen's kappa considers imbalance in class distribution and can. Therefore, g-mean and kappa can be most suitable for unbalanced datasets. The error rate is calculated for each class using the following equation:

$$\%error_{class\_j} = \frac{error_{class\_j}}{\sum_{i=0}^{1} error_{class\_i}} \times 100 \tag{3}$$

4 Results and Discussion

This section details the experimental study's results, illustrating the comparative effectiveness of the different data sampling techniques on the classification task.

4.1 Case Study 1

This study investigates the performance of different data sampling methods on a legal text classification task using a dataset comprising lawsuit decisions from the São Paulo Court. We particularly focus on the theme 0929, which involve compensation laws and financial corrections in savings accounts, respectively, under the Brazilian Consumer Protection Code [10].

The training set included 2,088 labelled as class_1 (theme 0929) and 11,485 as class_0 (other themes). The validation set consisted of 584 as class_1 and 1,976 as class_0. The production set contained only two from class_1.

We evaluated various sampling methods:

- Random Undersampling achieved a 2:1 ratio by reducing class_0 size by 36%.
- Near Miss Undersampling drastically reduced class_0 to a 1:12 ratio.
- SMOTE, Random Oversampling, and Borderline-SMOTE balanced the classes to a 1:1 ratio.
- ADASYN also balanced the classes to approximately a 1:1 ratio.
- Edited Nearest Neighbours reduced class_0 by 9%, achieving a 1:2 ratio.
- Tomek Links Undersampling was unable to balance the training set effectively.

First, we report the metrics results in every method at Table 2. After, we compare the available metrics for every method in Table 3.

Table 2: Performance comparison - Validation set

		Support	Prec	Mavg	Wavg	Rec	Mavg	Wavg	F1	Mavg	Wavg	Acc
Unbalance	C 0	1976	0.99	0.99	1.00	0.99	1.00	1.00	0.99	1.00	1.00	1.00
	C 1	584	0.97			0.97			0.97			
RU	C 0	1976	0.99	0.97	0.98	0.98	0.98	0.98	0.99	0.98	0.98	0.98
	C 1	584	0.95			0.98			0.96			
NMU	C 0	1976	1.00	0.85	0.93	0.88	0.94	0.91	0.94	0.88	0.91	0.91
	C 1	584	0.71			1.00			0.83			
SMOTE	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99
	C 1	584	0.97			0.96			0.97			
ADASYN	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99
	C 1	584	0.97			0.97			0.97			
RO	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99
	C 1	584	0.96			0.98			0.97			
Borderline	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99
	C 1	584	0.97			0.96			0.97			
ENN	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99
	C 1	584	0.96			0.97			0.97			
TLU	C 0	1976	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.99	0.98	0.99
	C 1	584	0.97			0.97			0.97			

Table 3: Metrics Comparison.

Method	Balanced accuracy	Precision	Recall	F1-score	G-mean	Kappa
Unbalance	0.997	0.997	0.997	0.997	0.997	0.991
RU	0.999	0.999	0.999	0.999	0.999	0.999
NMU	0.996	0.999	0.999	0.999	0.996	0.993
SMOTE	0.997	0.997	0.997	0.997	0.997	0.995
ADASYN	0.997	0.997	0.997	0.997	0.997	0.994
RO	0.996	0.996	0.996	0.996	0.996	0.993
Borderline	0.997	0.997	0.997	0.997	0.997	0.995
ENN	0.981	0.986	0.985	0.985	0.981	0.960
TLU	0.981	0.986	0.986	0.986	0.981	0.962

Initial results indicated that all methods performed robustly, with an f1-score around 97%. However, NMU showed diminished precision at 71%, highlighting the challenges of extreme undersampling.

Table 3 shows that RU had the highest score in g-mean e kappa metrics, achieving 0.99%. Figures 1 and 2 present the error rates across methods, illustrating variances in performance and highlighting the efficiency of balanced sampling techniques.

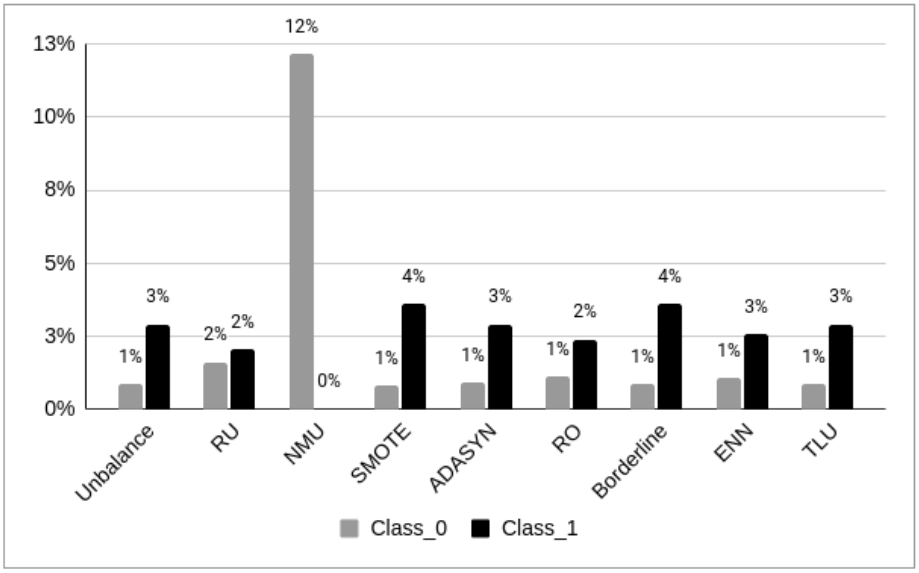

Fig. 1: Error Rate for Validation Dataset

4.2 Case Study 2

In the second experiment, we explore theme 1101, which focuses on establishing the final timeframe for accumulating remunerative interest in cases involving collective and individual legal actions. These actions aim to obtain compensation for omitting inflationary adjustments in savings accounts.

The training set included 1,421 labelled as class_1 (theme 1101) and 12,149 as class_0 (other themes). The validation set consisted 733 as class_1 and 1,827 as class_0. The production set contained 350 texts, none of class_1.

It is noted that the dataset sizes differ slightly between Case Study 1 (13,573 texts) and Case Study 2 (13,570 texts). This discrepancy is due to specific data filtering steps in Case Study 2, where three records were excluded during preprocessing because they did not meet the criteria for inclusion. This minor difference does not affect the analysis and conclusions drawn from the case studies.

We evaluated various sampling methods:

– Random Undersampling achieved a 1:1 ratio by reducing class_0 size by 12%.

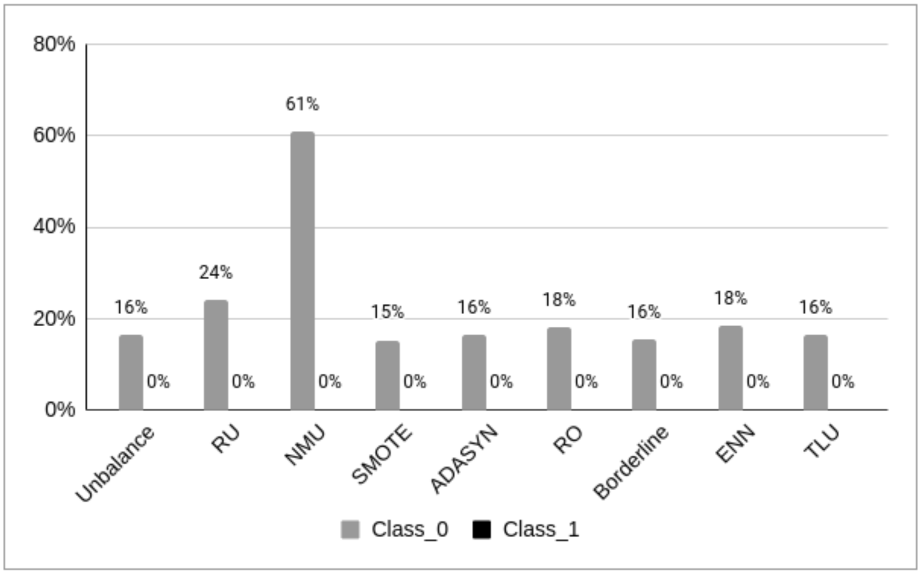

Fig. 2: Error Rate for Production Dataset

- Near Miss Undersampling reduced class_0 to a 1:11 ratio by reducing class_0 size by 1.3%.
- SMOTE, Random Oversampling, and Borderline-SMOTE balanced the classes to a 1:1 ratio.
- ADASYN also balanced the classes to approximately a 1:1 ratio.
- Edited Nearest Neighbours reduced class_0 by 2%, achieving a 1:8 ratio.
- Tomek Links Undersampling was unable to balance the training set effectively, reducing class_0 size by 99%.

The initial step of our analysis involves presenting the performance metrics obtained from each data sampling method, which are systematically compiled in Table 4. Subsequently, we perform a comprehensive comparison of these metrics across all methods, the results of which are detailed in Table 5.

Based on the Table 4, methods that maintain a balance between precision and recall, such as SMOTE and ADASYN, might be preferable in scenarios where both types of errors have similar costs.

Table 5 shows that RU had the highest score in all metrics, achieving 0.99%. The Unbalance and NMU methods show lower scores in balanced accuracy, kappa, and gmean compared to other methods, indicating less effectiveness in handling class imbalances. SMOTE and ADASYN, while not the top performers, show good consistency across metrics, suggesting they offer a reliable improvement over unbalanced datasets but might be slightly outperformed by techniques that involve more sophisticated data manipulation like ENN and TLU. Figures 3 and 4 show the poor performance of SMOTE and ADASYN in

Table 4: Performance comparison - Validation set

		Support	Prec	Mavg	Wavg	Rec	Mavg	Wavg	F1	Mavg	Wavg	Acc
Unbalance	C 0	1827	0.95	0.96	0.96	0.99	0.93	0.95	0.97	0.95	0.94	0.95
	C 1	733	0.98			0.86			0.91			
RU	C 0	1827	0.99	0.94	0.96	0.95	0.97	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.89			0.99			0.94			
NMU	C 0	1827	1.00	0.91	0.95	0.91	0.95	0.93	0.95	0.92	0.93	0.93
	C 1	733	0.81			1.00			0.89			
SMOTE	C 0	1827	0.97	0.95	0.96	0.97	0.95	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.93			0.92			0.93			
ADASYN	C 0	1827	0.96	0.95	0.95	0.98	0.94	0.96	0.97	0.94	0.95	0.96
	C 1	733	0.95			0.89			0.92			
RO	C 0	1827	0.99	0.94	0.96	0.95	0.97	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.89			0.99			0.94			
Borderline	C 0	1827	0.95	0.96	0.96	0.99	0.93	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.96			0.88			0.92			
ENN	C 0	1827	0.96	0.96	0.96	0.98	0.94	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.96			0.91			0.93			
TLU	C 0	1827	0.95	0.97	0.96	0.99	0.94	0.96	0.97	0.95	0.96	0.96
	C 1	733	0.98			0.88			0.93			

Table 5: Metrics Comparison.

Method	**Balanced accuracy**	**Precision**	**Recall**	**F1-score**	**G-mean**	**Kappa**
Unbalance	0.925	0.955	0.953	0.9952	0.922	0.883
RU	0.969	0.964	0.961	0.961	0.968	0.907
NMU	0.952	0.945	0.932	0.934	0.951	0.846
SMOTE	0.948	0.959	0.959	0.959	0.948	0.900
ADASYN	0.935	0.955	0.955	0.955	0.934	0.888
RO	0.969	0.964	0.960	0962	0.968	0.907
Borderline	0.935	0.957	0.957	0.956	0.933	0.892
ENN	0.944	0.961	0.961	0.961	0.944	0.904
TLU	0.935	0.961	0.960	0.960	0.933	0.898

validation and production data. NMU had a low error rate in validation but had a 22% error rate for class_0 in production data. The analysis indicates that while simple resampling methods can improve the balance and general performance of classification models, more sophisticated methods that focus on the quality of the sampled dataset can provide superior performance, especially in handling

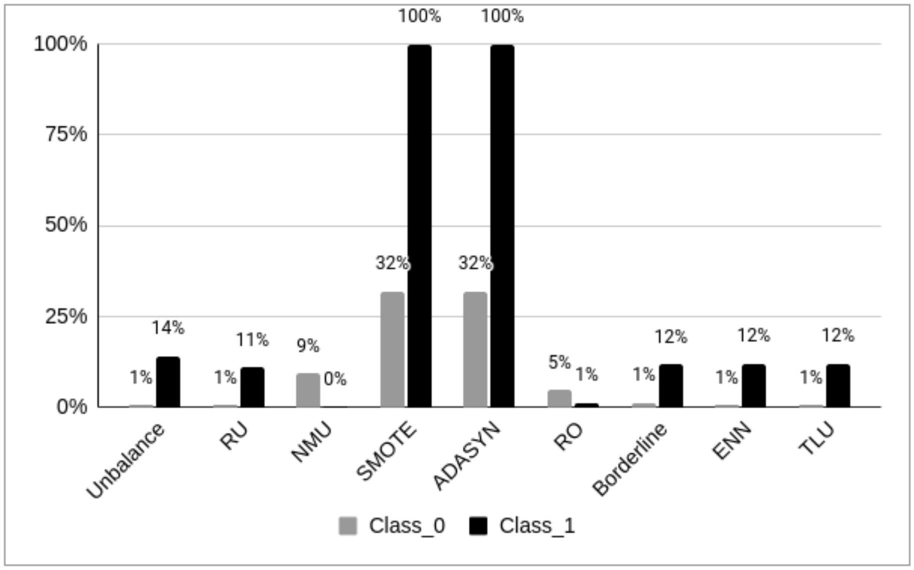

Fig. 3: Error Rate for Validation Dataset

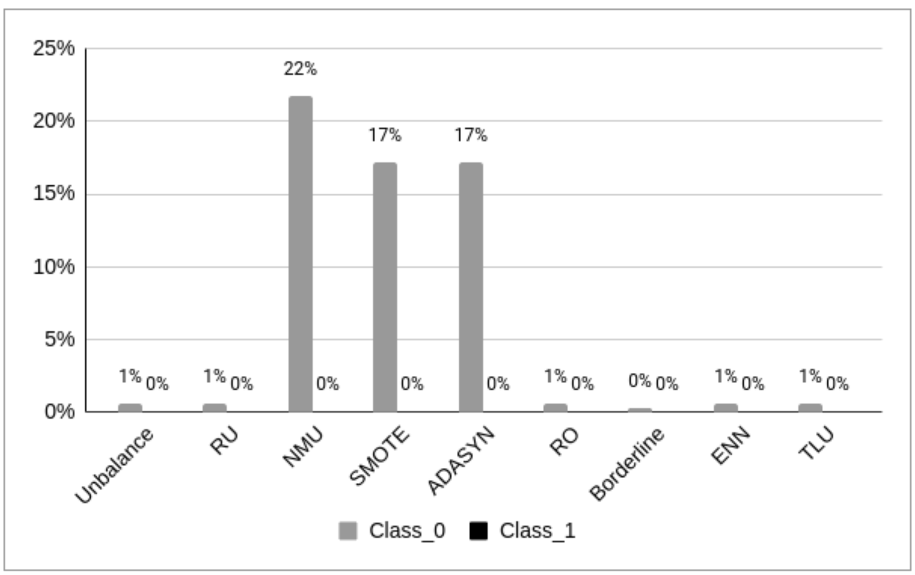

Fig. 4: Error Rate for Production Dataset

complex imbalances inherent in real-world datasets like those found in legal text classification.

5 Conclusion

This study addressed the challenge of imbalanced class distributions in machine learning, especially in legal datasets where minority class bias can affect model accuracy. The research found that techniques such as SMOTE and Borderline-SMOTE were effective in improving the representation of minority classes, improving overall model performance. Additionally, evaluation metrics such as geometric mean and Cohen's kappa were crucial for balanced performance assessment, helping avoid misleading results. Although this approach offers practical benefits in legal document classification, it also highlighted limitations: some sampling methods are computationally costly for large datasets, and oversampling can lead to overfitting with small minority classes. Future research is encouraged to explore more efficient sampling methods and broader metrics to enhance the robustness and scalability of the model in legal contexts. This work contributes to fairer algorithmic decision-making in the justice system by reducing bias and improving classification accuracy in imbalanced data scenarios.

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